Week_4_Milestone 3: White-Paper
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Milestone 3: White-Paper

Title: Customer Segmentation to Improve Marketing Efficiency and Campaign Targeting

Business Problem

Many businesses use broad marketing strategies that treat all customers the same, even though customers differ in income, spending habits, purchasing frequency, and responsiveness to campaigns. This limits the effectiveness of promotions and may cause companies to waste marketing resources on customers who are unlikely to respond. The business problem addressed in this project is how to identify meaningful customer groups using historical customer data so that a company can better target marketing efforts, improve retention, and increase campaign efficiency. By segmenting customers into groups with similar purchasing and behavioral characteristics, businesses can create more personalized and cost effective strategies.

Background and History

Customer segmentation is a widely used technique in business analytics and marketing. The goal is to divide a customer base into smaller groups that share similar characteristics, allowing businesses to better understand who their customers are and how they behave. Rather than marketing to everyone in the same way, firms can use segmentation to tailor promotions, communication, and product recommendations. In data science, clustering is one of the most common methods used for segmentation because it identifies patterns in data without needing predefined labels. K means clustering is especially popular because it is relatively simple, efficient, and effective for grouping customers with similar numeric characteristics. For this project, customer segmentation was used to analyze income, spending, purchases, and engagement metrics in order to identify actionable customer groups.

Data Explanation

The dataset used in this project was a customer segmentation dataset containing demographic, household, and purchasing information. The original dataset included variables such as birth year, education, marital status, income, household composition, recency of purchases, spending by product category, purchase behavior across channels,

and campaign response history. Several new variables were created during data preparation, including Total Spending, Total Purchases, Age, Total Children, and Customer Tenure Days. These derived variables made it easier to summarize customer behavior and compare segments. Records with unrealistic ages were removed, and missing income values were handled during preprocessing. The final dataset was suitable for clustering because it contained multiple behavioral and financial indicators that could be standardized and compared across customers.

Methods

This project used descriptive analysis, feature engineering, data standardization, K means clustering, and visualization. First, the dataset was explored using summary statistics and charts to understand customer spending and purchasing patterns. Next, derived features were created to better reflect overall customer value and engagement. The clustering analysis focused on variables such as income, recency, total spending, total purchases, web visits, age, and number of children in the household. These variables were scaled using StandardScaler so that no single measure would dominate the clustering process. The elbow method and silhouette score were used to evaluate cluster quality and support the selection of four clusters. Principal component analysis was then used to visualize the clusters in two dimensions.

Analysis

The analysis identified four distinct customer segments. Cluster 1 represented the highest value customers, with an average income of 78165.75, average spending of 1341.26, and the highest campaign response rate at 0.24. These customers also had the fewest monthly web visits, suggesting they may be efficient buyers who already know what they want. Cluster 3 represented another strong segment, with average income of 61218.34 and average spending of 895.84. This cluster also showed high purchase activity and moderate response rates. Cluster 0 and Cluster 2 were lower value groups. Cluster 0 had average income of 36938.64, very high recency at 74.98, low spending of 133.22, and very low response at 0.04. Cluster 2 had the lowest income at 34818.09 and similarly low spending at 121.08, but showed somewhat better response behavior at 0.16. Category level spending profiles showed that Cluster 1 spent the most across all product categories, especially wines and meat products, while Cluster 3 ranked second. These results suggest that the business has at least two high opportunity segments and two lower value segments requiring different engagement strategies.

Conclusion

The findings show that customer segmentation can provide meaningful business insight by identifying groups with different levels of value, engagement, and marketing responsiveness. The most important result is that not all customers should be approached in the same way. High value customers can be retained through premium offers and

loyalty strategies, while lower value segments may require re engagement or more cost conscious promotions. Overall, the analysis demonstrates that clustering can support smarter marketing decisions and improve how companies allocate resources.

Assumptions

This analysis assumed that the dataset is a reasonable representation of customer behavior and that the available variables capture important differences in purchasing patterns. It also assumed that K means clustering is appropriate for this type of data and that four clusters provide a useful balance between statistical separation and business interpretability. Another assumption is that historical customer behavior can help inform future marketing strategy.

Limitations

This project has several limitations. The dataset does not include all possible factors influencing customer behavior, such as geographic variation, seasonality, product preferences over time, or competitor actions. The data appears to represent a snapshot rather than a full longitudinal customer history, which limits the ability to analyze change over time. In addition, K means clustering requires the number of clusters to be chosen in advance and can be sensitive to scaling and outliers.

Challenges

One challenge in this project was selecting the most relevant variables for clustering. Including too many variables can reduce interpretability, while including too few may oversimplify customer differences. Another challenge was ensuring that the final clusters were not only statistically separate but also useful from a business perspective. Balancing mathematical output with real world interpretability was an important part of the project.

Future Uses and Additional Applications

This project could be extended by using additional customer data such as geographic region, online browsing history, promotion exposure, or product level transaction details. Future analyses could compare segmentation results over time or test whether the same clusters remain stable across multiple campaigns. The model could also be expanded using other machine learning methods such as hierarchical clustering or predictive modeling for customer response and retention.

Recommendations

Based on the results, the business should prioritize Cluster 1 and Cluster 3 for personalized marketing and retention strategies because these groups generate the highest spending and strongest purchase activity. Cluster 0 may benefit from re engagement campaigns because of its high recency and low response rate. Cluster 2 may respond better to value based promotions because it has lower income and low overall spending but a somewhat better response rate than Cluster 0. Marketing resources should be

allocated differently across these groups instead of using a one size fits all campaign approach.

Implementation Plan

A practical implementation plan would begin with assigning each active customer to a segment based on the clustering model. The marketing team could then design different campaigns for each segment, such as loyalty rewards for high value customers and discount driven outreach for lower value groups. Dashboards should be created to monitor spending, response rates, and retention by segment. The segmentation model should be reviewed and refreshed on a regular basis so that it continues to reflect current customer behavior. Finally, campaign performance should be measured using A B testing to confirm that segmentation based strategies improve outcomes.

Ethical Considerations

Customer segmentation raises important ethical concerns. Even when direct identifiers are removed, customer data still reflects personal behavior and should be handled responsibly. Privacy protection is essential, and customer information should only be used for legitimate business purposes. Another concern is bias in interpretation. If segments are described in ways that unfairly stereotype people based on income, age, family structure, or purchasing power, the analysis could reinforce harmful assumptions. Businesses must be careful not to use segmentation to unfairly exclude groups from opportunities or manipulate vulnerable customers. In this project, segmentation should be viewed as a tool for improving customer experience and marketing efficiency, not for discrimination.

Appendix A - Cluster Summary

Cluster 0 is a low income, low spending, and low response segment with high recency. Cluster 1 is the highest value segment, with the highest income, total spending, and campaign response. Cluster 2 is another lower value segment with lower income and spending but slightly better response than Cluster 0. Cluster 3 is a strong middle to upper tier segment with high spending and high purchase activity. Category level results show that Cluster 1 leads spending across all product categories and Cluster 3 ranks second.

References

Dapat, V. (n.d.). Customer segmentation clustering. Kaggle.

Data Science with Onur. (2025, July 19). Customer segmentation using machine learning K means clustering full Python data science project [Video]. YouTube.

Appendix A. Supporting Documentation

This appendix provides supporting documentation for the customer segmentation analysis conducted in this project. The purpose of the appendix is to summarize the data preparation steps, derived variables, clustering inputs, and final outputs used to support the findings presented in the white paper.

Dataset Overview

The dataset used for this project contains customer demographic, purchasing, and campaign response information. Variables include income, education, marital status, number of children in the household, enrollment date, recency, product spending, purchase channels, website visits, and campaign response indicators.

Data Preparation

Several preprocessing steps were performed before clustering. Missing values were removed from the dataset to ensure a clean analytical sample. The **Dt_Customer** field was converted to datetime format for time-based calculations. Unreasonable age values were filtered out by calculating age from year of birth and restricting the sample to adult customers within a realistic range.

Feature Engineering

New variables were created to improve the analysis. **Total_Spending** was calculated as the sum of spending across product categories, including wine, fruits, meat products, fish products, sweet products, and gold products. **Total_Purchases** was calculated as the combined number of web, catalog, and store purchases. **Age** was derived from year of birth. **Customer_Since** was created from the customer enrollment date, and **Total_Children** was calculated as the sum of **Kidhome** and **Teenhome**.

Variables Used for Clustering

The clustering analysis was performed using the following variables:

- Income
- Recency
- Total_Spending
- NumWebPurchases
- NumCatalogPurchases
- NumStorePurchases
- NumWebVisitsMonth

These variables were selected because they reflect both customer value and customer engagement across purchasing channels.

Standardization and Model Selection

Because the clustering variables were measured on different scales, the data was standardized using **StandardScaler** before fitting the model. The elbow method and silhouette score were used to compare different cluster solutions. Based on these diagnostics and the interpretability of the results, a four-cluster solution was selected.

Final Cluster Outputs

The final analysis produced a cluster assignment for each customer and generated summary tables showing the average characteristics of each cluster. These outputs included:

- **customer_segmentation_clustered_output.csv**
- **cluster_profile_summary.csv**
- **cluster_category_profile.csv**

Interpretation Support

The cluster summaries showed meaningful differences in income, spending, purchases, web activity, family structure, and response behavior. In particular, one segment had the highest average income and spending, while another high-performing segment showed strong purchasing activity with somewhat different browsing behavior. Lower-spending clusters showed lower purchase volume and weaker overall customer value.

Visual Support Included in the White Paper

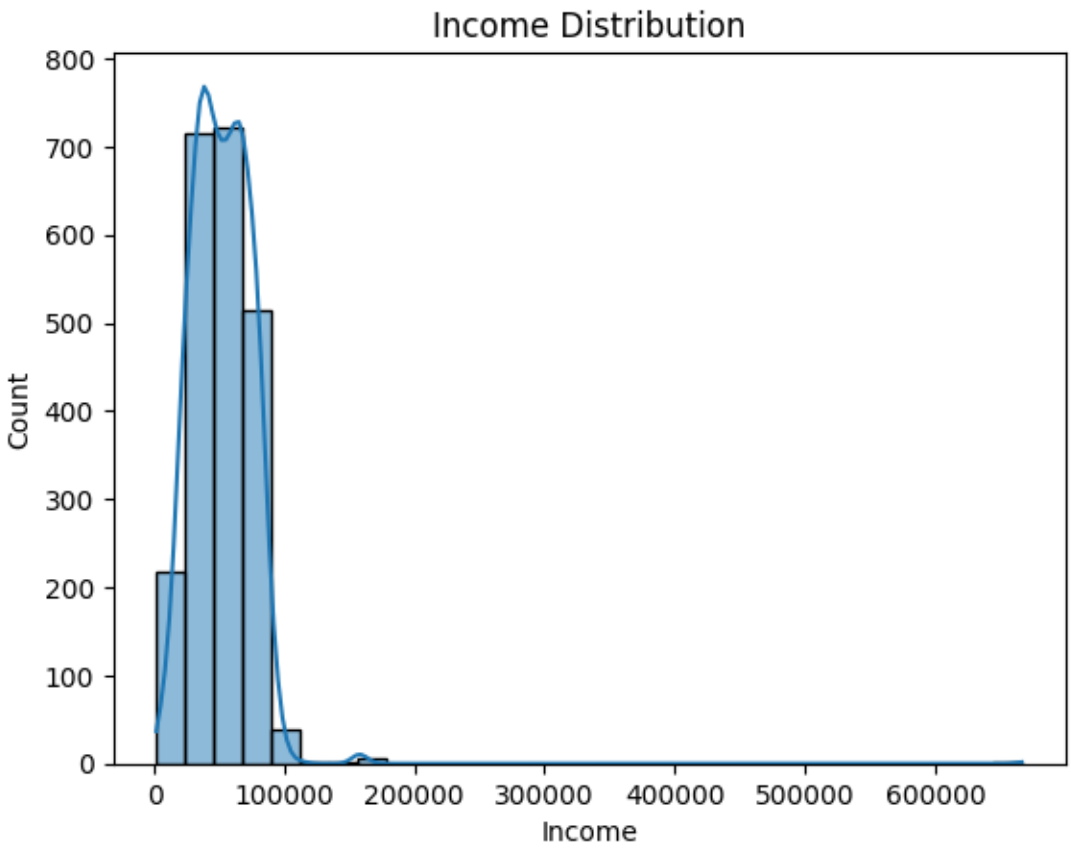
The white paper includes several illustrations created from the analysis, such as:

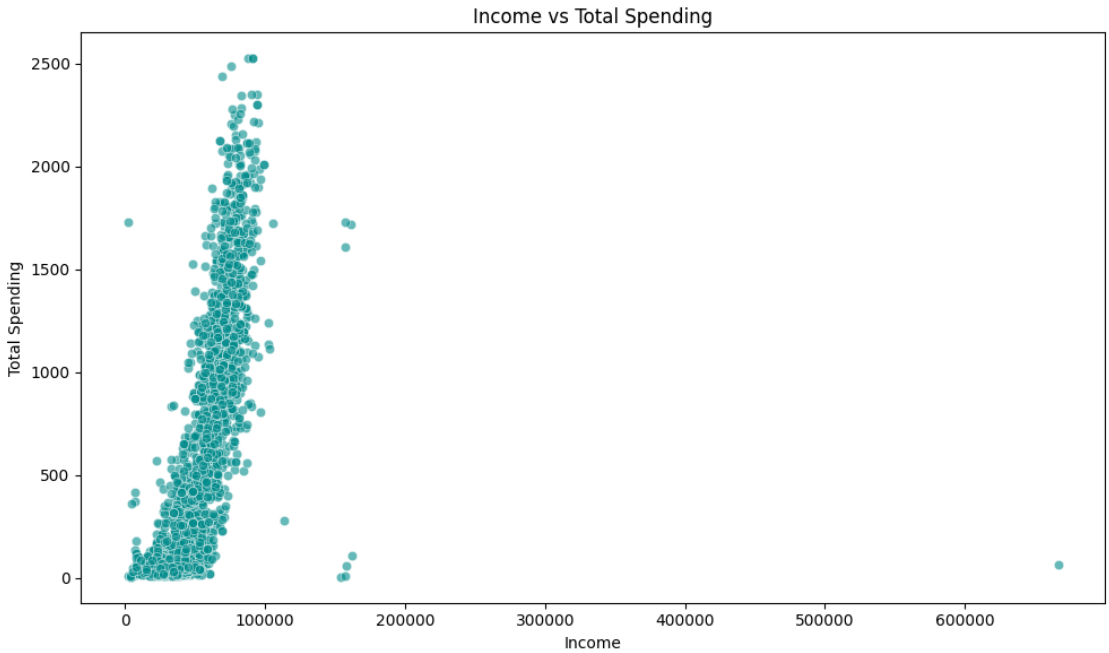
- Income distribution
- Total spending distribution
- Income versus total spending scatterplot
- Elbow method plot
- Average total spending by cluster
- Average campaign response rate by cluster
- PCA cluster visualization

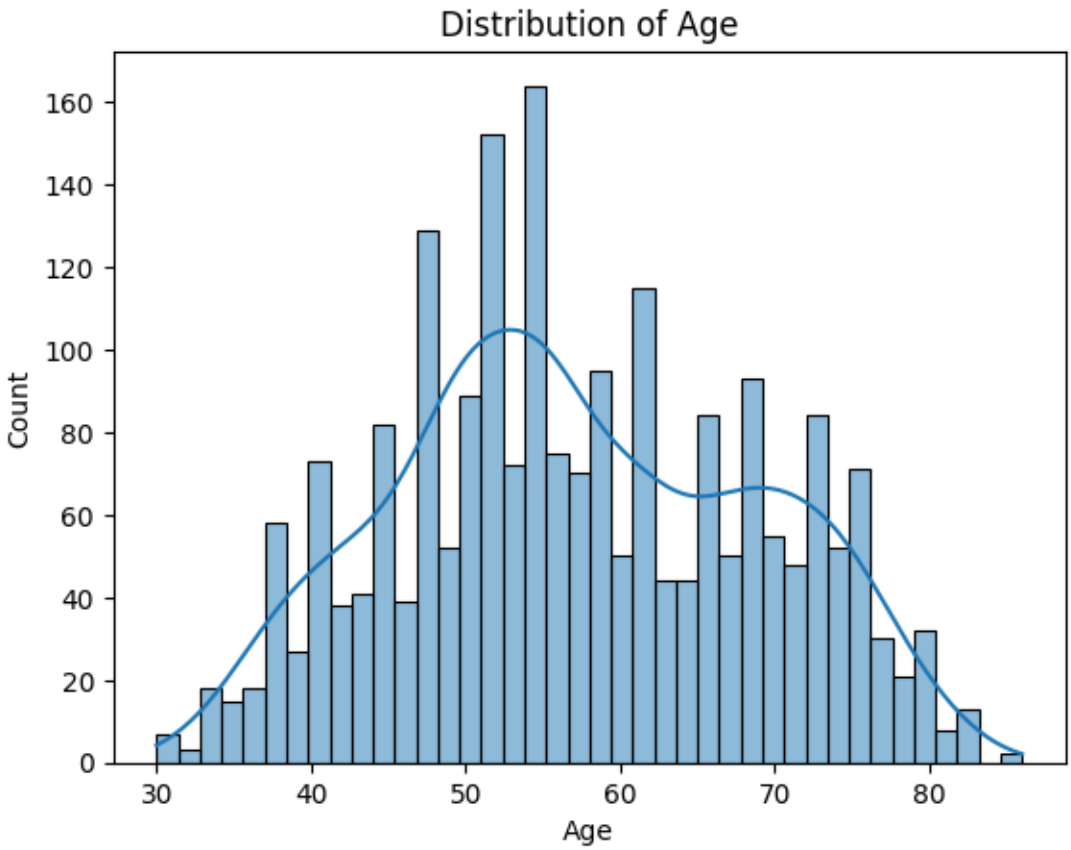
These visuals were used to support interpretation of the customer segments and demonstrate how the final clustering solution was chosen.

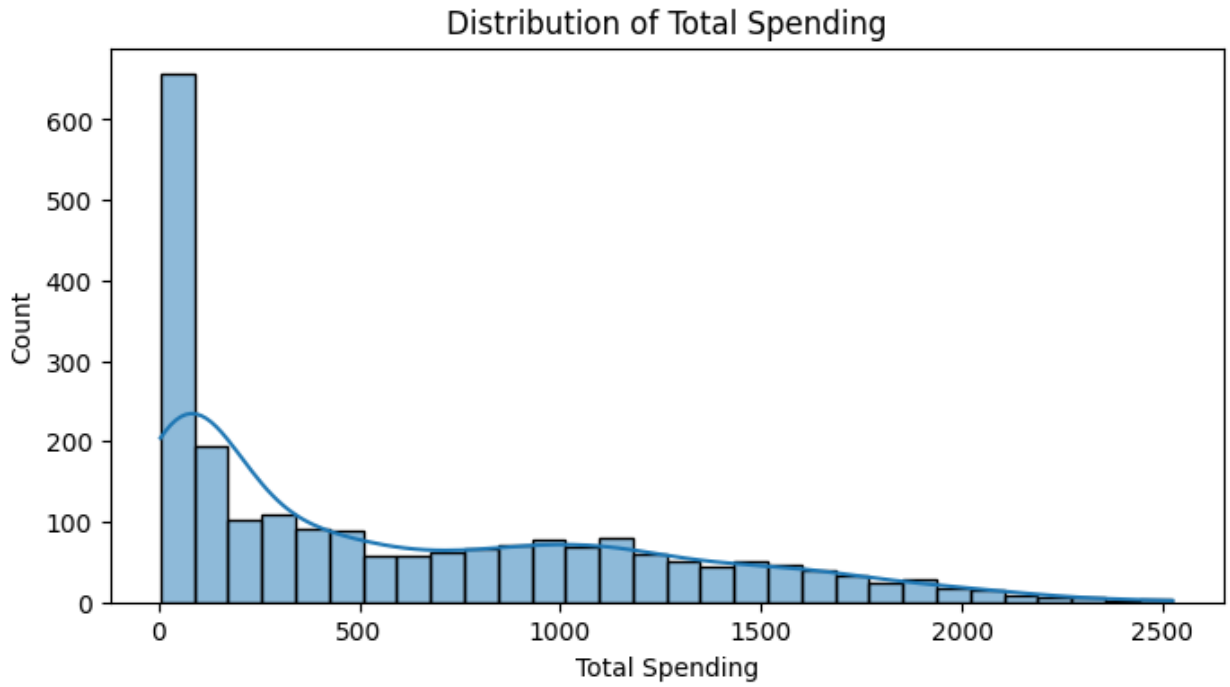
Visuals

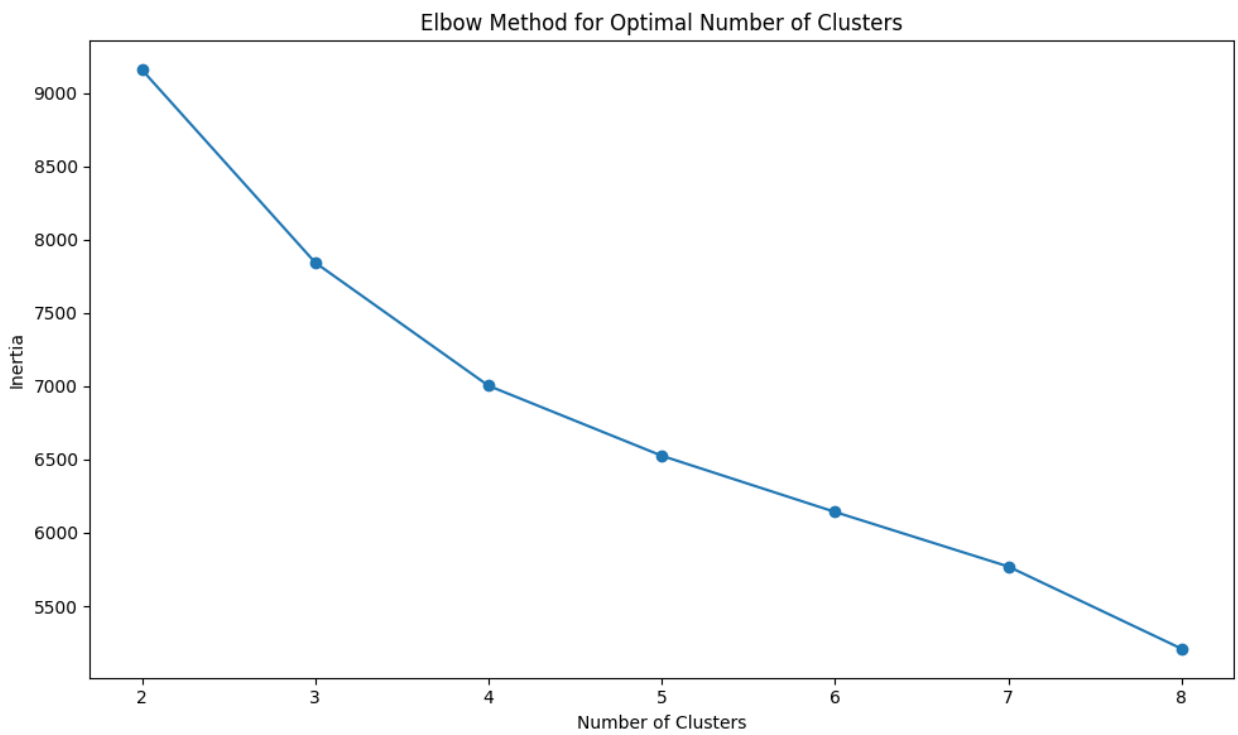
The visualizations in this project were used to support both the exploratory analysis and the final interpretation of the customer segments. Early visuals, including the income distribution, total spending distribution, age distribution, and the scatterplot of income versus total spending, helped reveal important patterns in customer value, purchasing behavior, and variation across the dataset. Additional clustering visuals, such as the elbow method, silhouette score comparison, average total spending by cluster, average campaign response rate by cluster, and the PCA cluster projection, helped justify the selection of four clusters and made the differences among customer groups easier to interpret. Together, these figures provide a clear visual foundation for the analysis and strengthen the connection between the data, the clustering results, and the final marketing recommendations.

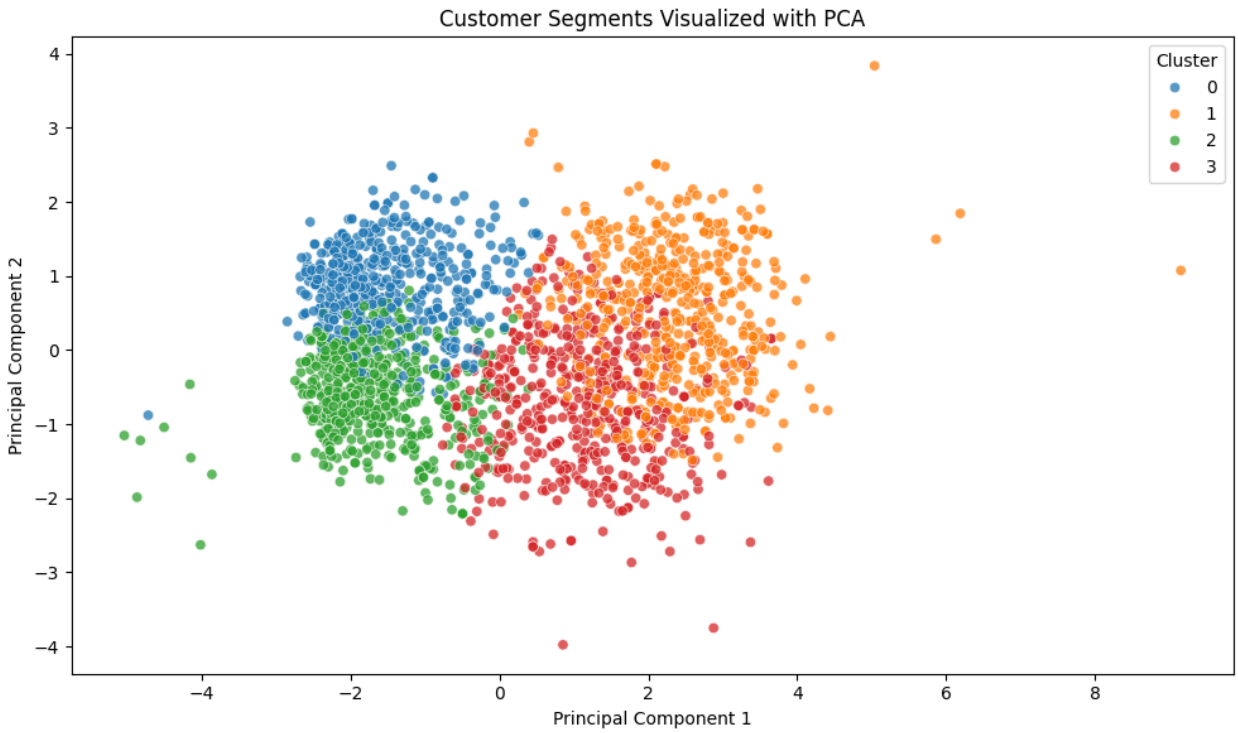


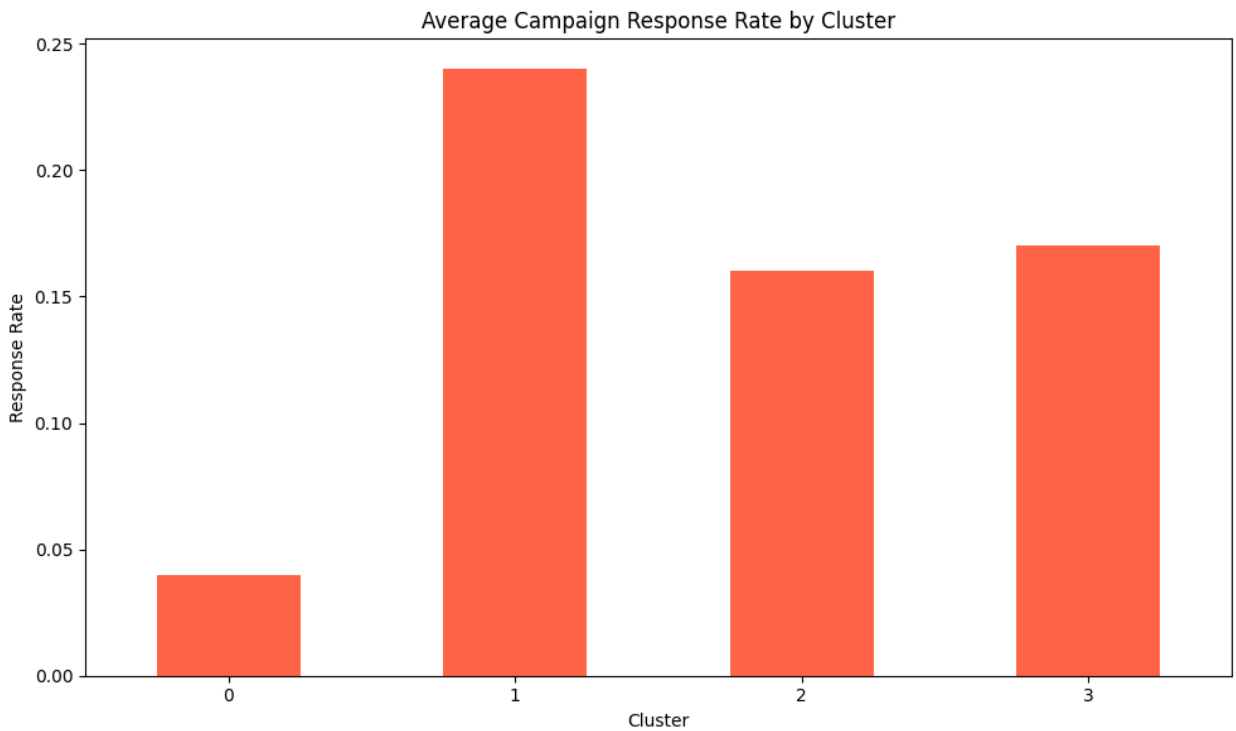

Income vs Total Spending
Total Spending
Income
0
500
1000
1500
2000
2500
0
100000
200000
300000
400000
500000
600000

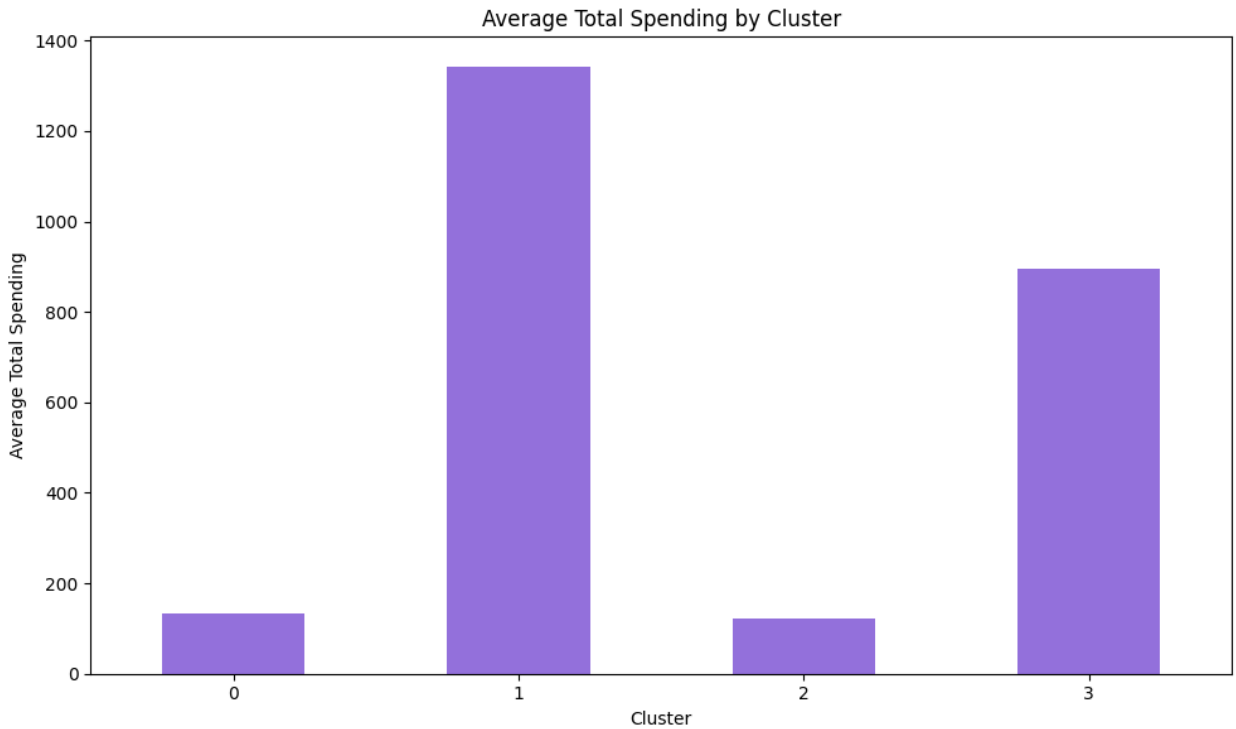

Distribution of Age
Count
Age
0
20
40
60
80
100
120
140
160
30
40
50
60
70
80

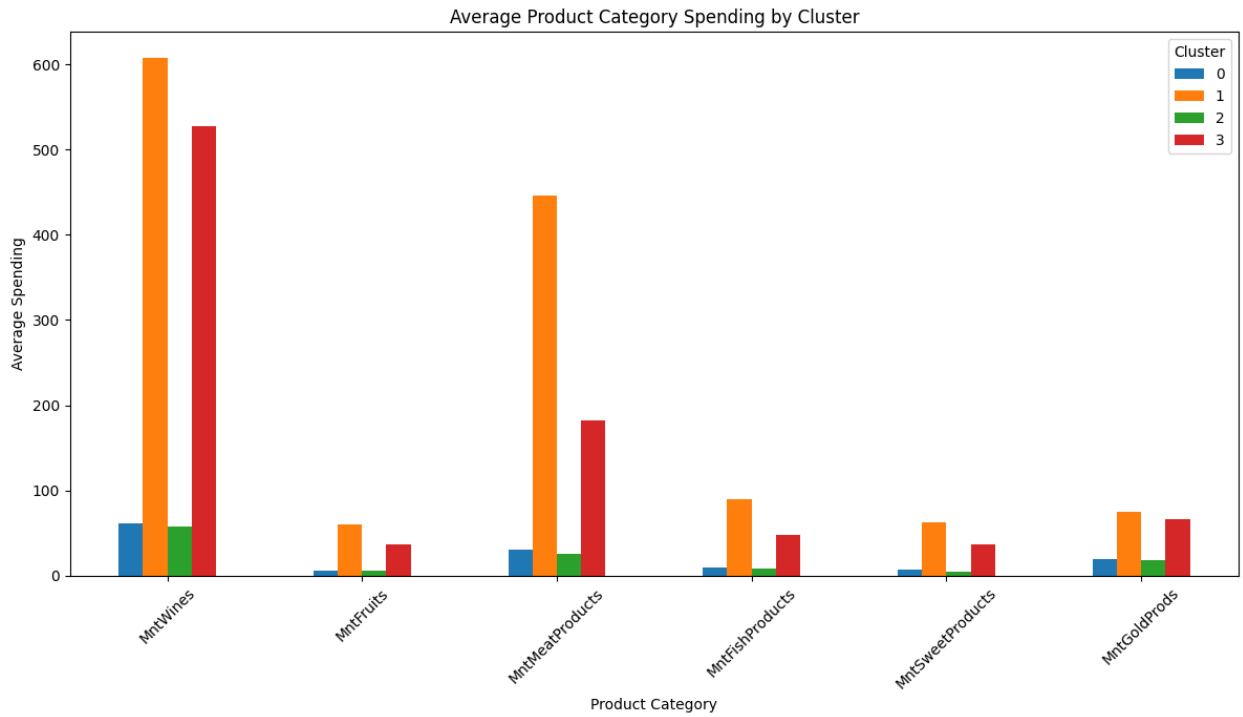

Distribution of Total Spending
Count
Total Spending
0
100
200
300
400
500
600
0
500
1000
1500
2000
2500


Elbow Method for Optimal Number of Clusters
Inertia
Number of Clusters
5500
6000
6500
7000
7500
8000
8500
9000
2
3
4
5
6
7
8


Customer Segments Visualized with PCA
Cluster
0
1
2
3
Principal Component 2
Principal Component 1
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
6
8


Average Campaign Response Rate by Cluster
Response Rate
Cluster
0.25
0.20
0.15
0.10
0.05
0.00
0
1
2
3


Average Total Spending by Cluster
Average Total Spending
Cluster
0
1
2
3
0
200
400
600
800
1000
1200
1400


Average Product Category Spending by Cluster
Average Spending
Product Category
Cluster
0
1
2
3
MntWines
MntFruits
MntMeatProducts
MntFishProducts
MntSweetProducts
MntGoldProds
0
100
200
300
400
500
600

Q&A

1. Why did you choose customer segmentation for this project?

I chose customer segmentation because it is a practical example of how data science can support business decision-making. Organizations collect massive quantities of customer data, but that information is only useful if the data can be turned into actionable insight. Customer segmentation helps reveal patterns in spending, purchasing behavior, and engagement, allowing businesses to better understand who their customers are and how to market to them more effectively.

2. What business problem does this project solve?

This project addresses the problem of inefficient marketing. Businesses use broad campaigns that treat all customers the same, even though customers differ in value, behavior, and responsiveness. As a result, companies may waste resources on low-performing campaigns or fail to focus on their most valuable customers. By identifying customer segments, this project helps support more targeted marketing strategies that can improve campaign effectiveness and resource allocation.

3. Why did you use K-means clustering?

I used K-means clustering because it is a well-established and interpretable unsupervised learning method for grouping similar observations. It is especially useful in segmentation problems where there are no predefined customer labels. In this project, K-means provided a straightforward way to cluster customers based on numerical variables related to value, behavior, and engagement, while also producing results that could be clearly explained from a business perspective.

4. How did you decide on four clusters?

The decision to use four clusters was based on a combination of statistical evaluation and interpretability. I used the elbow method to examine how model inertia changed across different numbers of clusters, and I used the silhouette score to evaluate how well the clusters were separated. Based on those results, four clusters provided a strong balance between model performance and practical usefulness, making the segmentation both meaningful and easy to interpret.

5. What variables were most important in creating the clusters?

The most important variables were those included in the clustering model because they captured customer value, purchasing activity, and engagement. These variables were

income, recency, total spending, number of web purchases, number of catalog purchases, number of store purchases, and number of monthly web visits. Together, these measures provided a broad picture of how much customers spend, how recently they interacted with the business, and which channels they use most often.

6. What did the clusters reveal about the customers?

The clusters revealed that the customer base was not uniform and that customers differed meaningfully across income, spending, channel activity, and engagement. One segment represented high-value customers with the highest spending and income levels. Another segment showed active buyers with strong purchasing behavior. The remaining groups reflected customers with lower spending, lower engagement, or more minimal purchasing activity. These findings suggest that different customer groups likely require different marketing strategies.

7. Which customer segment is the most valuable?

The most valuable segment is the high-value customer group, which had the highest average income and total spending in the analysis. This group also showed strong purchasing activity and greater overall customer value than the other segments. From a marketing perspective, this segment would be the top priority for retention efforts, loyalty programs, and premium offers because these customers already contribute the greatest business value.

8. How can a business use these clustering results?

A business can use these results to create more targeted and effective marketing strategies. High-value customers can be prioritized for loyalty programs, retention campaigns, and personalized premium offers. Mid-value or active customers can be targeted with upselling opportunities and customized promotions to increase lifetime value. Lower-engagement customers can be approached with reminders, discounts, or reactivation campaigns. Overall, the clustering results allow a business to align its marketing strategy with the behavior and value of each customer group instead of using a one-size-fits-all approach.

9. What are the limitations of this analysis?

This analysis has several limitations. First, the results depend on the variables available in the dataset, which means the clusters only reflect the specific customer characteristics that were measured. Second, K-means requires the number of clusters to be chosen in advance, so the solution depends partly on model selection decisions. Third, clustering simplifies customer behavior into groups even though customer behavior is often more

complex and may change over time. Finally, the results were not tested against real campaign outcomes, so additional validation would be needed to confirm how well the segments perform in practice.

10. What would you improve if you continued this project?

If I continued this project, I would improve it by comparing K-means with other clustering methods such as hierarchical clustering or DBSCAN to determine whether similar segments emerge. I would also incorporate additional variables, such as customer tenure, geographic region, or detailed campaign history, if those data were available. Another important improvement would be validating the segments through A/B testing or campaign performance analysis to see whether segment-based targeting leads to better results. Finally, I would explore ways to update the segmentation over time so it could become an ongoing business tool rather than a one-time analysis.